

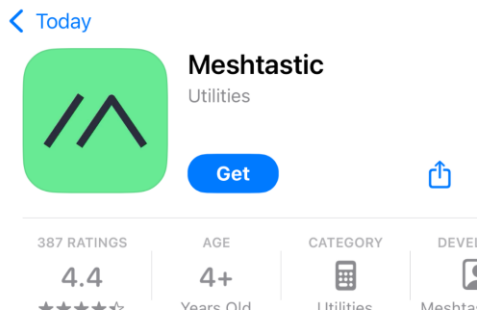
Basic User - Operator

Installing Meshtastic Phone App

- Refer to [Meshtastic software documentation](#) for additional support on installing the mobile apps.

Apple

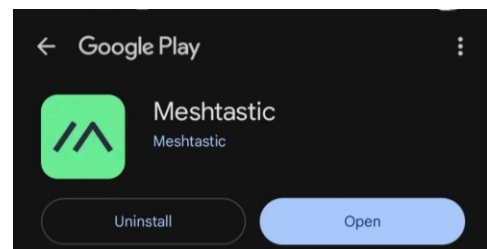
- Download [Meshtastic](#) from Apple Store or scan QR code below



- Open Meshtastic app once installation is complete
- Click **"Allow"** when prompted to find Bluetooth devices
- Click **"Get started"**
- Click **"Configure notification permissions"** and press **"Allow"** when prompted to allow notifications
- Click **"Allow"** when prompted about sending Critical Alerts
- Click **"Configure location settings"** and press **"Don't allow"** as location is not needed in-cave

Android

- Download [Meshtastic](#) from Google Play store, through [alternate repo\(s\)](#), or scan the QR code below:



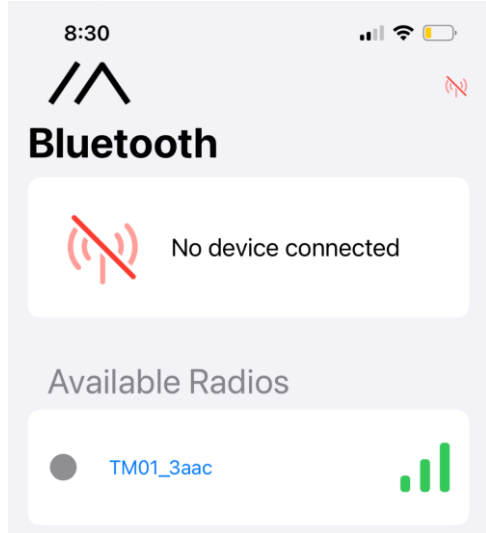
- Open Meshtastic app once installation is complete
- Press **"Next"** 2x and **"Done"** to get through intro
- Click **"Scan"** under Bluetooth and press **"Allow"** when prompted about finding nearby devices
- If prompted, allow notification permission

Connecting to Known Node

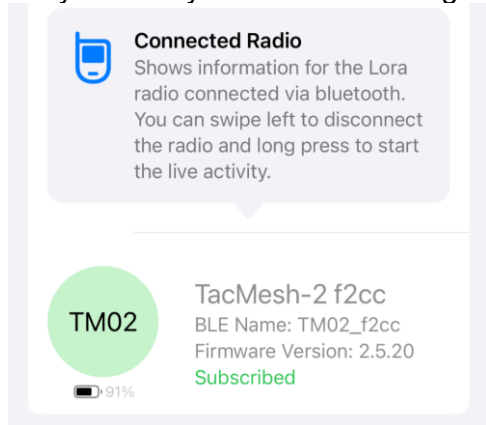
- Assuming radio is powered up and have the app open

Apple

- On the **"Bluetooth"** tab under **"Available Radios,"** select the desired node name

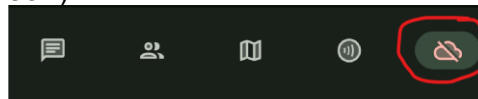


- Verify node says **"Subscribed"** in green text

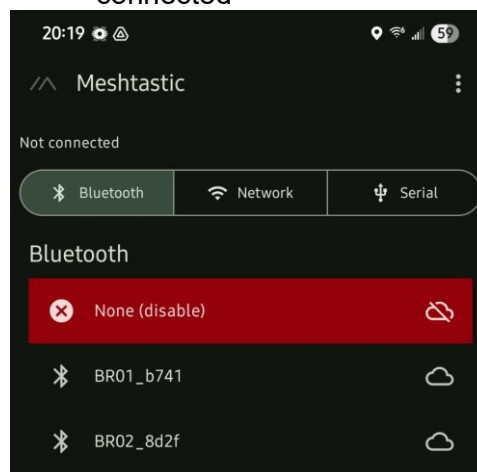


Android

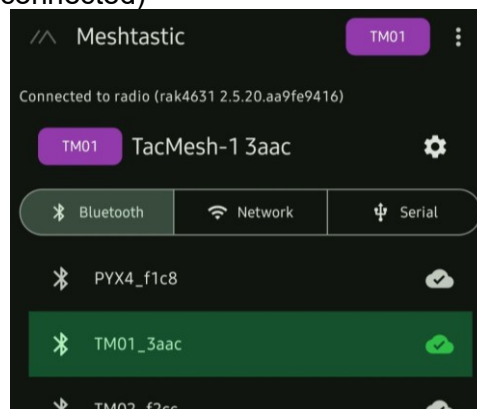
- Click Connections tab in bottom right (**cloud icon**)



- Under the **"Bluetooth"** list, select the desired node's name
 - Cloud will be red with slash when not connected



- Verify cloud icon turns green (indicates connected)

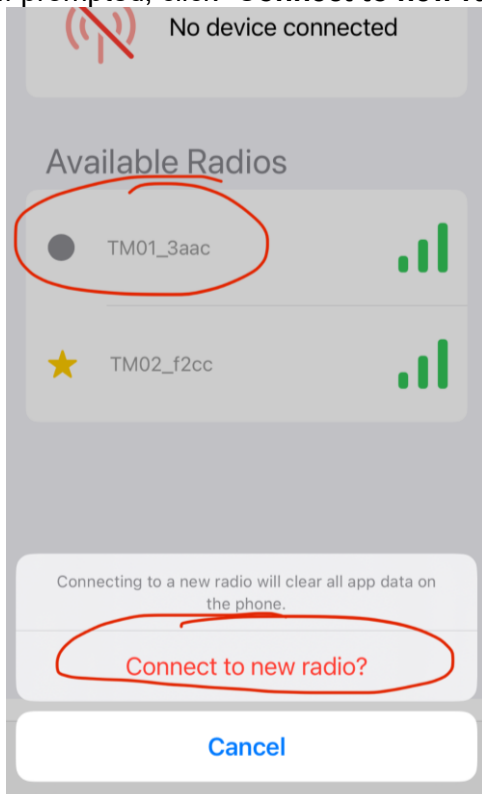


Pairing to New Node

- Assuming radio is powered up and have the app open

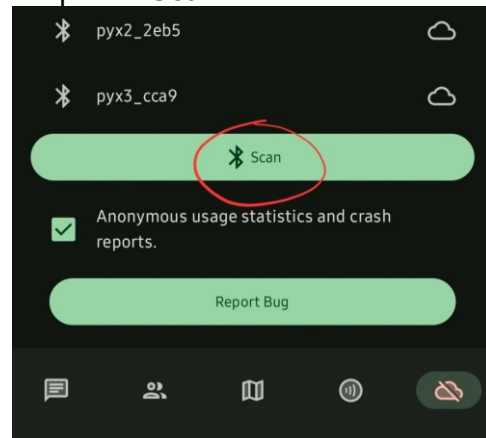
Apple

- Under “**Available Radios**,” select the node name
- If prompted, click “**Connect to new radio?**”



Android

- Click Connections tab in bottom right (**cloud icon**)
- Under the “**Bluetooth**” list, scroll to bottom and press “**Scan**”



- When prompted, enter the Bluetooth PIN “**911911**” and press “**Pair**”
- Verify node says “**Subscribed**” in green text

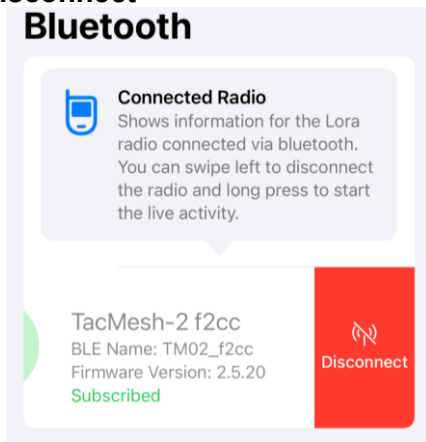
- Under “**Select a Bluetooth device**,” click on new node
- When prompted, enter the Bluetooth PIN “**911911**” and press “**Pair**”
- Verify settings load and cloud icon goes green

Disconnecting from a Radio

- Assuming radio is powered up and you are connected to a radio via the app
- NOTE:
 - It is crucial to **ALWAYS disconnect from a node when walking away from it**. Staying connected will lock out all other users from being able to pair to it, even from a long distance.

Apple

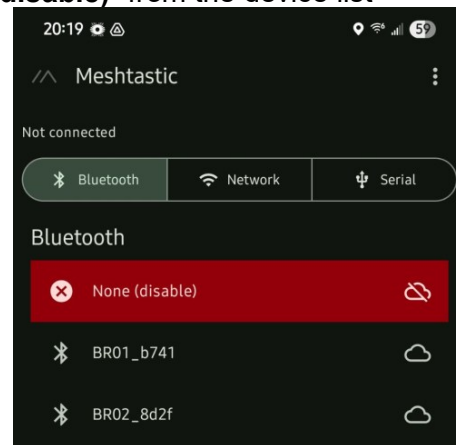
1. Swipe **left** on the Subscribed node and press **"Disconnect"**



2. Verify app says **"No device connected"**

Android

1. On the Connections tab, select **"None (disable)"** from the device list



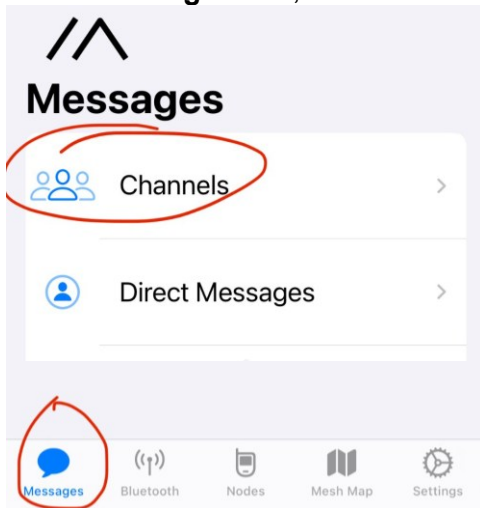
2. Verify cloud icon turns red and **"None (disable)"** is highlighted

Sending Message to Chatroom

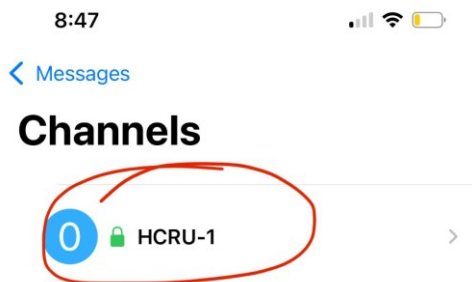
- Chatroom messages are visible to anyone on the network
- Assuming radios are powered up and you are connected to a radio via the app
- NOTE:
 - **“Acknowledged”** indicates some neighboring node has picked up your message. This does not mean it has been delivered but that it has entered the network.
 - **DO NOT send the “Alert bell” in any messages.** This may be reserved for a special function for setup mode for FLAMINGO.

Apple

1. Under **“Messages”** tab, select **“Channels”**

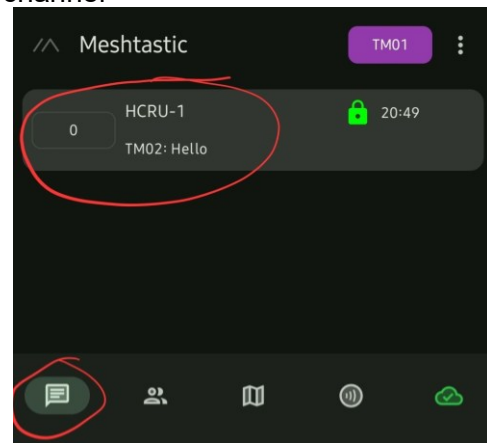


2. Under **“Channels,”** select **“HCRU-1”** or relevant channel



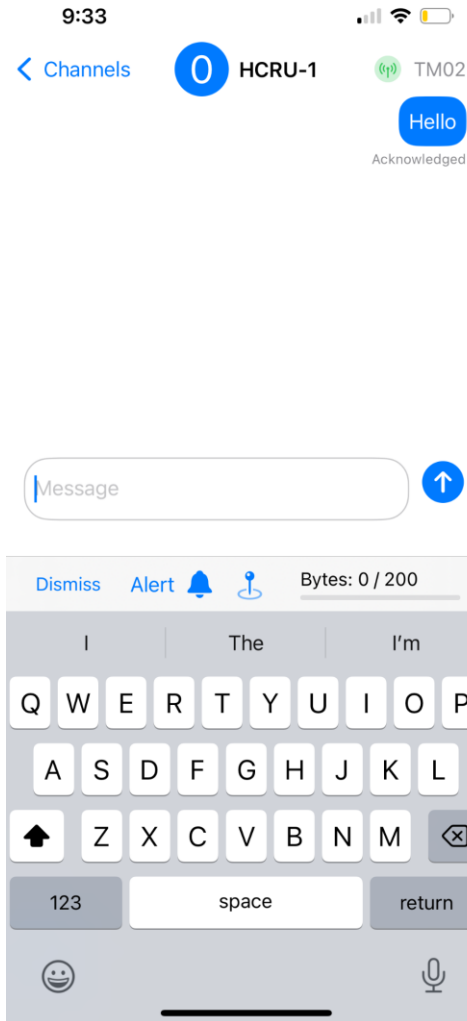
Android

1. Select the Messages tab in the bottom left (**chat bubble icon**)
2. Select the **“HCRU-1”** chatroom or relevant channel



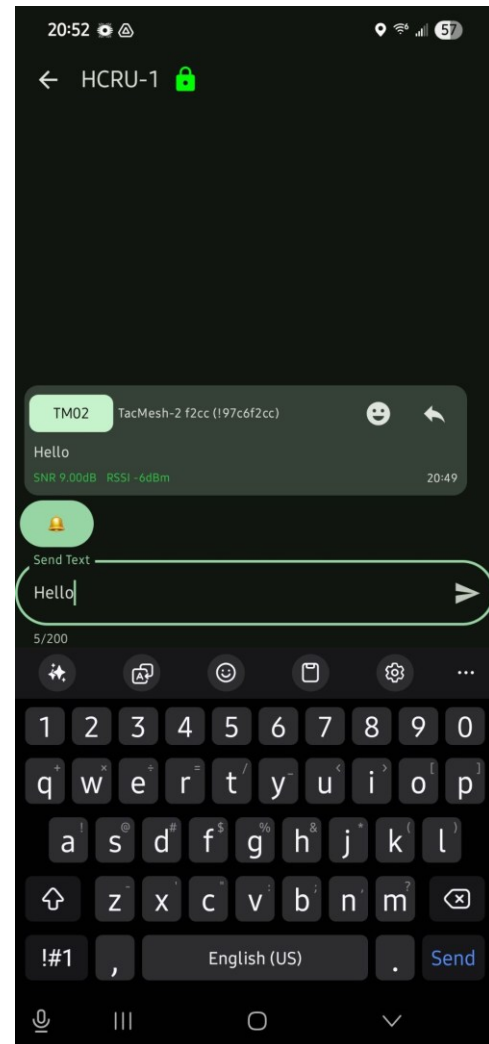
Apple

3. Compose message and press send



Android

3. Compose and send message(s)

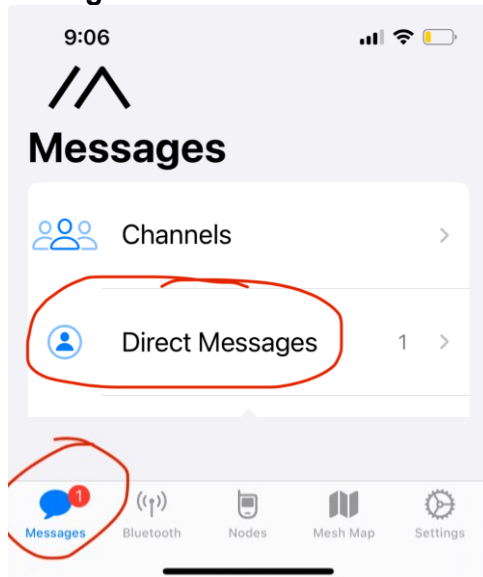


Sending Direct Messages

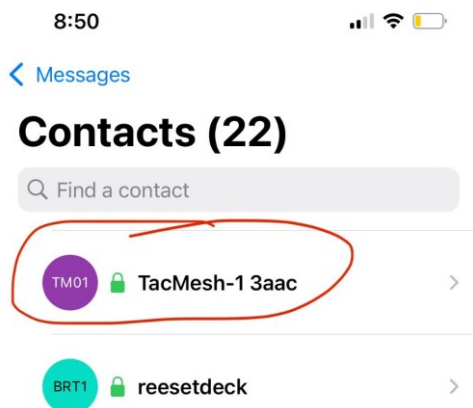
- Direct messages are only visible to the recipient and yourself
- Assuming radios are powered up and you are connected to a radio via the app
- NOTE:
 - **“Acknowledged”** indicates some neighboring node has picked up your message. This does not mean it has been delivered but that it has entered the network.
 - **“Delivery Confirmed”** (user and checkmark icon) indicates that the recipient’s node has received your message and is digitally acknowledging
 - **DO NOT send the “Alert bell” in any messages.** This may be reserved for a special function for setup mode for FLAMINGO.

Apple

1. Under **“Messages”** tab, select **“Direct Messages”**

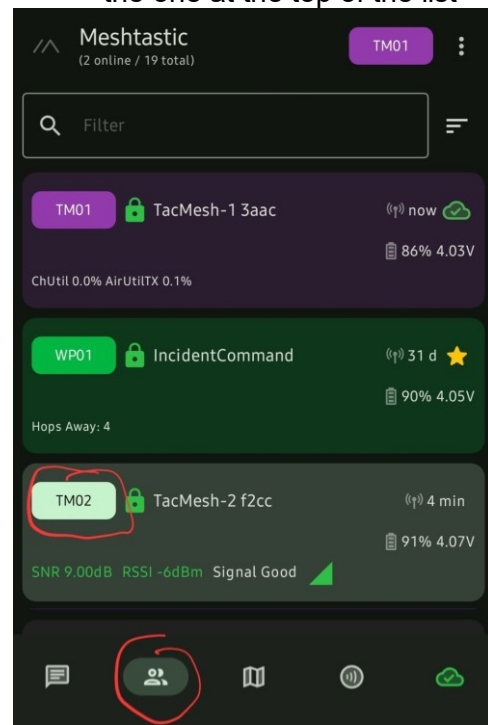


2. Under **“Contacts,”** select the recipient



Android

1. Select the Contacts tab in the bottom left (**people icon**)
2. Click on the recipient’s **callsign** from the list
 - Note your own node will always be the one at the top of the list



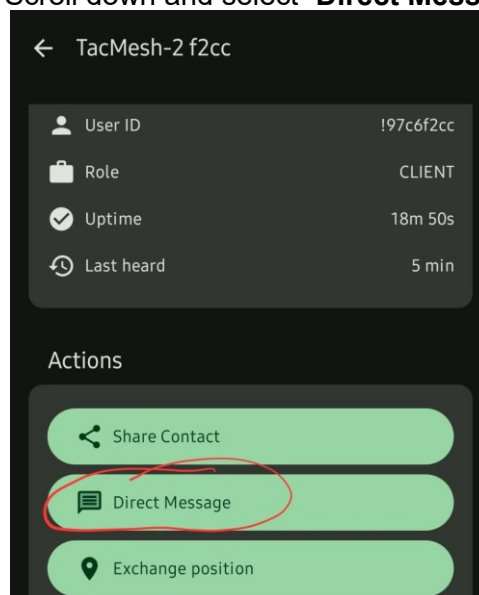
- 3.

Apple

3. Compose message and press send

Android

4. Scroll down and select **“Direct Message”**



5. Compose message and press send

Intermediate User – Setup Technician

Network Setup

See introduction video here (www.youtube.com/watch?v=R3LtLcnrpAk) for an overview of Flamingo and an example of setting up a mixed network.

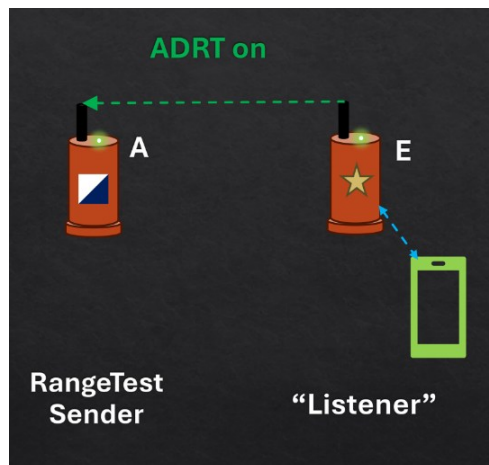
Preparation

Before setting up a network it is important to consider the following factors:

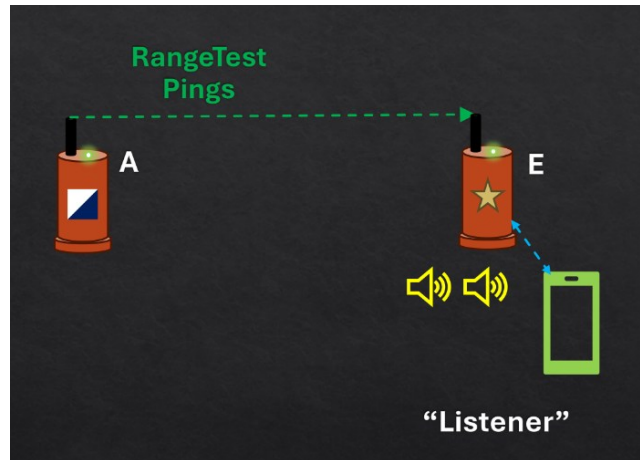
1. Number of personnel
 - . Evaluate the number of personnel available for the task of setting up comms
 - . Consider the relative skills of the comms task force and ensure the Listener Node operator has been trained (or at least oriented) in these procedures
 - . Set all personnel-assigned nodes (that are not part of the network backbone) to **CLIENT_MUTE** role
2. Cave layout
 - . Ideally the general path of the terrain is known before entering the cave
 - . If not known, have the hasty or initial search team(s) assess the path to determine if there are:
 - . Winding corners
 - . Low ceilings/crawls
 - . Squeezes
 - . Breakdowns
 - . If these tight conditions exist, consider using Hybrid nodes and laying phone wire through these features instead of only RF nodes
3. Remember that RF nodes can also be used above ground to span distance between IC and the cave entrance(s)
4. Number and types of nodes available
5. Once resources and path terrain are more clear, determine a notional network topology (layout) and prepare and assign comms equipment as needed
6. Consider the cave entrance(s) and designate an appropriate node for the transition from surface to underground
7. Set the node with the best line-of-sight (LOS) possible between the surface node and the entrance passage
8. Ensure the entrance node is secure and clear from traffic entering and exiting the cave (i.e. make sure it can't easily get knocked down a pit)
9. If the entrance is tight and/or immediately begins with tight passage or turns, consider starting at the entrance with a Hybrid node

Deploying Network

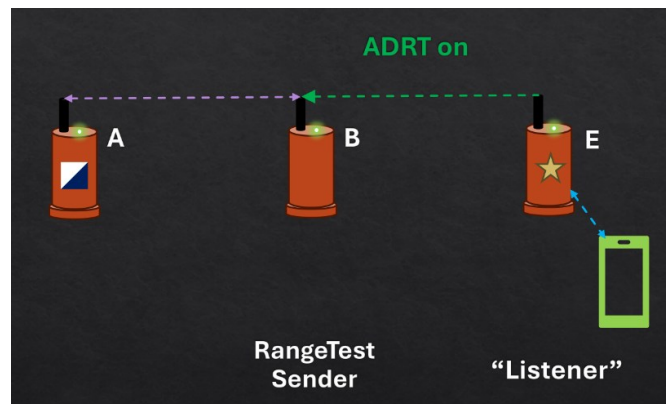
1. Identify Incident Command (IC) location and establish primary surface node
 - . This node should be at or near Incident Command
2. Power on primary node and connect to IC mobile device
3. [OPTIONAL] connect IC node to a laptop over USB-C for logging:
 - . Open VSCode (or similar)
 - . Identify the COM port
 - . Open a Serial Monitor and set baud to **9600**
 - . Start logging
4. Designate mobile RF node as the “Listener” node
5. On the Listener RF node, send either (1) the “**Alert Bell character**” button or (2) the message “**ADRT on**” in the Meshtastic app as a *Direct Message* to the IC node
 - WARNING:** DO NOT send “Alert Bell” character into any *chatrooms* as it will not activate setup mode



6. The Listener node will now receive signal strength updates (“pings”) from the IC radio at 15 second intervals
7. Use the auto/visual haptics on the Listener node to travel along intended network path and evaluate signal
 - . 1 beep indicates Good signal
 - . 2 beeps indicates Marginal signal
 - . 3 beeps indicates Low signal
 - . When the Listener node first indicates Low signal, search for a suitable spot nearby in the passage that has Marginal (or possibly Low, if necessary) signal
 - . Ideally the Average SNR should be **≥3dB** otherwise reliability will be degraded



8. Place a *new* RF node at that location and power it on
NOTE: Do not place the Listener node down, hold onto it for setup
9. Send (1) another **"Alert Bell character"** command or (2) the message **"ADRT off"** directly to the IC radio and verify that the pings *stop*
10. Send a message to IC from the Listener node notifying of the new node placement and thereby verify comms
11. Send an **"Alert Bell character"/"ADRT on"** command now to the new RF node and continue forward into the passage



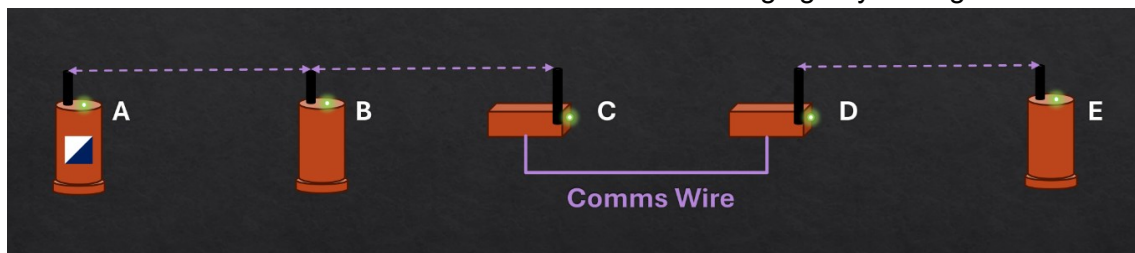
12. Repeat this process by "pinging" the previous RF node and leap-frog forward
13. When network is complete, stop setup mode by turning off pings at the last node
14. Optionally annotate a cave map with node and wire placement if available

Setting up Hybrid Node

1. Use similar approach to previous section but *no range tests are needed*
2. Unfold the Hybrid radio antenna and orient it upward
3. Set the node flat on the ground
4. Clean and/or strip last ½" (1cm) of both strands of one end of comms wire
NOTE: Keep track of wire polarity! We recommend coloring light and dark flags on each conductor so they are not accidentally crossed during setup.
5. Match polarity and clip wire ends into the corresponding alligator clips on the node
6. Cover exposed clips/wire ends with electrical tape to prevent accidental contact
7. Unspool wire and anchor/tack nearby wire to keep node from getting moved

8. Unspool wire and keep it out of the main path
9. Repeat node setup on far end

Hybrid and RF nodes can be mixed in the same network without changing any settings.



Tearing Down Network

1. Power off nodes in reverse order and repackage them
2. Coil up and stow wires
3. [Optional] export "rangetest.csv" log from IC node

Advanced User – Developer

Defer to the Flamingo Github repo for the most up-to-date procedures.

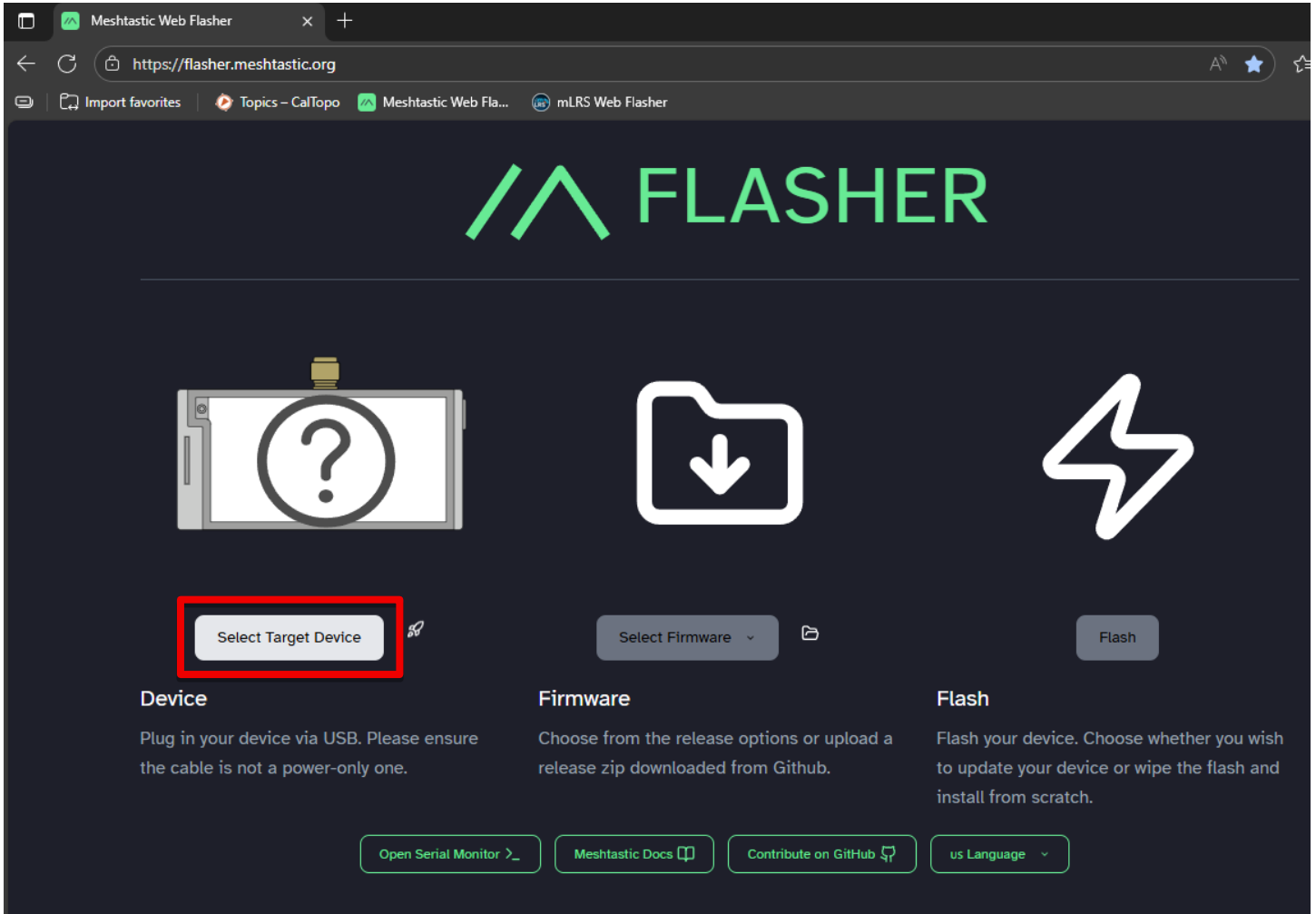
Installing Meshtastic CLI

To install the Meshtastic Python Command Line Interface (CLI), refer to the official documentation [here](#).

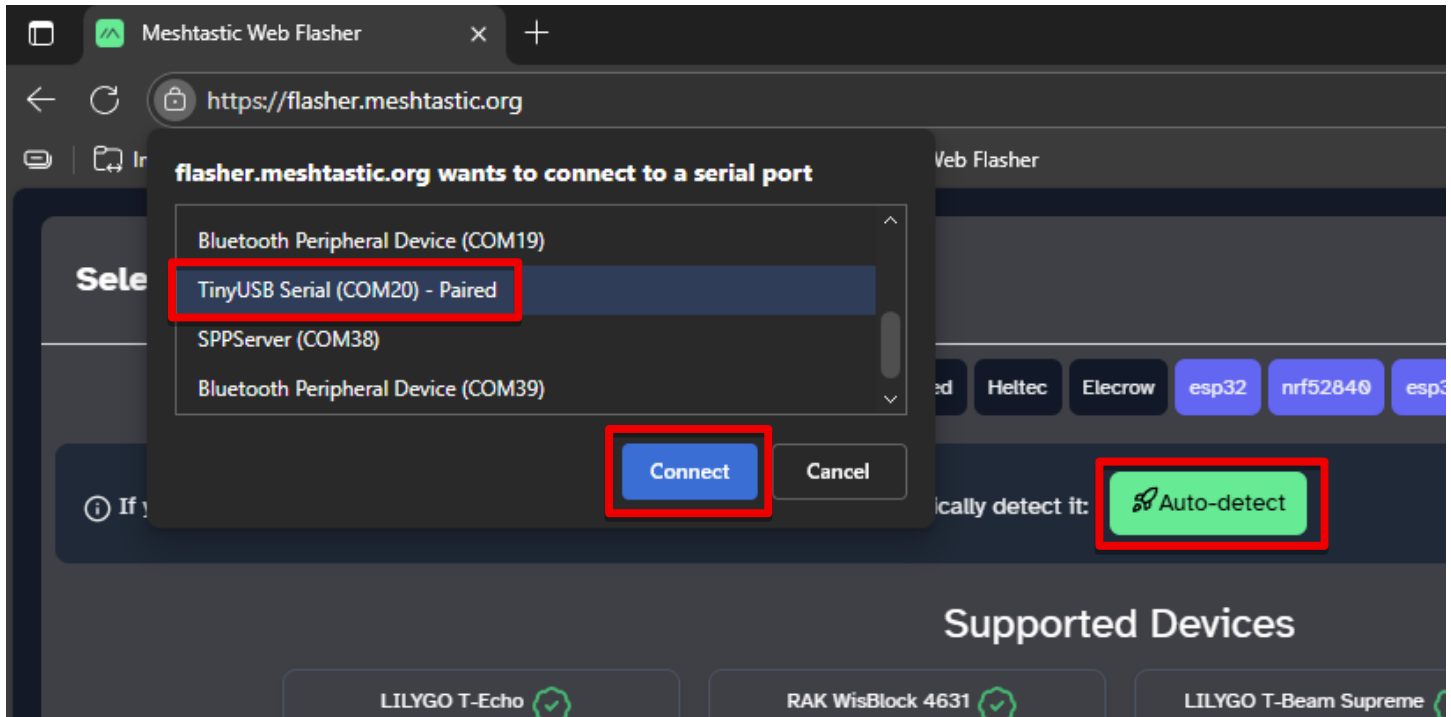
Updating Firmware (Legacy)

Windows PC - Browser

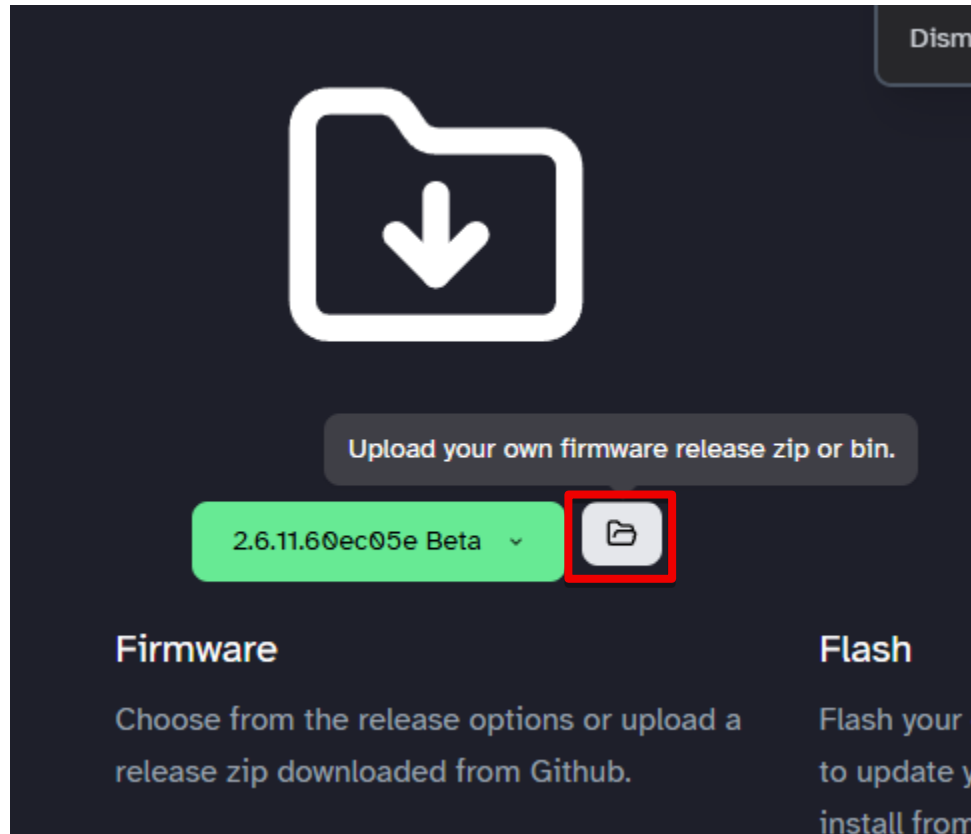
1. Clone latest branch from github.com/rbressems/flamingo and navigate to “firmware” folder
2. Identify appropriate FLAMINGO firmware (*.uf2 file) for your radio
 - . Verify the firmware filename matches the target (e.g. cavegen2 or bridge)
3. Open [Meshtastic web flasher](#) in *any browser but Firefox*
4. Plug in radio over USB *data* cable
5. Click "Select Target Device"



6. Click "Auto-detect"
7. Select "**TinyUSB Serial**" connection
8. Verify COM port in Windows Device Manager if needed



9. If radio automatically enters DFU mode (i.e. displays "Device successfully entered DFU mode" message and Explorer opens to the external drive), skip to Step 15
10. Press folder icon to "Upload your own firmware release"
11. Navigate to FLAMINGO FW and select it



12. Press Flash
13. Press "Enter DFU Mode"
14. Select same device from list and confirm
15. Verify RAK is mounted (i.e. files are visible in Windows Explorer)
16. Click "**Download UF2**" file
17. Copy appropriate FLAMINGO *.uf2 file into RAK (DFU) drive
18. Device will reboot automatically upon transfer completion
19. Ignore errors about diskette error if toward very end of transfer

Command Line Interface

1. Open command line (e.g. Command Prompt, Powershell, or Git Bash)
2. Plug radio into PC using USB-C data cable
3. [Optional] Update CLI using command "**pip install --upgrade meshtastic**" if on Linux or
4. Enter command "**meshtastic --enter-dfu**"
5. Copy appropriate FLAMINGO *.uf2 firmware into DFU folder root directory

Flashing Radio Settings

When configuring multiple radios, it is crucial that all radios have the same critical settings. Radios can easily become inoperable when settings are manually changed. Properly tracking and managing radios using the setup script is the best way to ensure all radios are synchronized and ready to go.

The *.YML file allows all the important radio settings to be captured in a single place and then pushed to all radios using that template.

```
settings:
  bluetooth.mode : "FIXED_PIN"
  bluetooth.fixed_pin: "911911"
  device.role : "CLIENT"
  lora.hop_limit: "31"
  lora.modem_preset: "MEDIUM_SLOW"
  lora.sx126x_rx_boosted_gain: "True"
  lora.tx_power: "30"
  lora.tx_enabled: "True"
  lora.ignore_mqtt: "True"
  lora.override_duty_cycle: "True"
  position.fixed_position: "False"
  device.rebroadcast_mode: "ALL"
  network.wifi_enabled: "False"
  telemetry.device_update_interval: "1800"
  range_test.enabled: "True"
  position.gps_mode: "NOT_PRESENT"
  serial.baud: "BAUD_4800"
```

Figure 0.1 Example of radio settings within *.YML file

Pushing YML File

1. Clone latest branch from FLAMINGO repo or navigate to “utils/configurator” folder
2. Open command line (e.g. Command Prompt, Powershell, or Git CMD) in the “**utils/configurator**” folder
3. [Optional] Update CLI using command “**pip install --upgrade meshtastic**”
4. Flash settings using the command “**python config.py --set <yml_file>**” or “**python configure_node_2.X.py <yml_file> -s**” where <yml_file> is the appropriate preconfigured *.YML from inside the “configs” folder
 - . Typically called “**cave_node_MMMYY.yml**”

NOTE: The radio may reboot a few times during this process. Wait until the CLI returns to normal input before unplugging radio.

Editing YML file

1. Open “cave_node_MMMYY.yml” or similar FLAMINGO *.YML template file with a text editor
2. Add or subtract lines inside the “settings:” section using the following syntax:
<section>.<setting>: “<value>”
 Example: Set radio transmission power to 30dBm would be **lora.tx_power: “30”**
3. Refer to CLI section and setting names as catalogued in the [Meshtastic Radio Configuration settings page](#)
4. If changing or adding chatrooms, replace the “**psk**” and corresponding “**name**” value fields in the “**channels**” section of the *.YML file.
5. Save *.YML file with descriptive name